

A Modified BPaL Regimen for Tuberculosis Treatment replaces Linezolid with Inhaled Spectinamides

Reviewed Preprint

Published from the original preprint after peer review and assessment by eLife.

About eLife's process

Reviewed preprint version 1

April 12, 2024 (this version)

Sent for peer review

January 23, 2024

Posted to preprint server

November 16, 2023

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Abstract

The Nix-TB clinical trial evaluated a new 6-month regimen containing three-oral-drugs; bedaquiline (B), pretomanid (Pa) and linezolid (L) (BPaL regimen) for treatment of tuberculosis (TB). This regimen achieved remarkable results as almost 90% of the multidrug resistant (MDR) or extensively drug resistant (XDR) TB participants were cured but many patients also developed severe adverse effects (AEs). The AEs were associated with the long-term administration of the protein synthesis inhibitor linezolid. Spectinamide 1599 (S) is also a protein synthesis inhibitor of *Mycobacterium tuberculosis* with an excellent safety profile but which lacks oral bioavailability. Here we hypothesize that inhaled spectinamide 1599, combined with BPa —BPaS regimen—has similar efficacy to that of BPaL regimen while simultaneously avoiding the L-associated AEs. The BPaL and BPaS regimens were compared in the Balb/c and C3HeB/FeJ murine chronic TB efficacy models. After 4-weeks of treatment, both regimens promoted equivalent bactericidal effect in both TB murine models. However, treatment with BPaL resulted in significant weight loss and the complete blood count suggested development of anemia. These effects were not similarly observed in mice treated with BPaS. BPaL treatment also decreased myeloid to erythroid ratio and increased concentration of proinflammatory cytokines in bone marrow compared to mice receiving BPaS regimen. During therapy both regimens improved the lung lesion burden, reduced neutrophil and cytotoxic T cells counts while increased the number of B and helper and regulatory T cells. These combined data suggest that inhaled spectinamide 1599 combined with BPa is an effective TB regimen that avoids L-associated AEs.

Importance

Tuberculosis (TB) is an airborne infectious disease that spreads via aerosols containing *Mycobacterium tuberculosis* (Mtb), the causative agent of TB. TB can be cured by administration of 3-4 drugs for 6-9 months but there are limited treatment options for patients infected with multidrug (MDR) and extensively resistant (XDR) strains of Mtb. BPaL is a new all-oral combination of drugs consisting of Bedaquiline (B), Pretomanid (Pa) and

Linezolid (L). This regimen was able to cure ~90% of MDR and XDR TB patients in clinical trials but many patients developed severe adverse effects (AEs) associated to the long-term administration of linezolid. We evaluated a new regimen in which Linezolid in the BPaL regimen was replaced with inhaled spectinamide 1599. In the current study, we demonstrate that 4-weeks of treatment with inhaled spectinamide 1599 in combination with Bedaquiline and Pretomanid has equivalent efficacy to the BPaL drug combination and avoids the L-associated-AEs.

eLife assessment

In this **useful** study, the authors report the efficacy, hematological effects, and inflammatory response of the BPaL regimen (containing bedaquiline, pretomanid, and linezolid) compared to a variation in which Linezolid is replaced with the preclinical development candidate spectinamide 1599, administered by inhalation in tuberculosis-infected mice. The authors provide **convincing** evidence that supports the replacement of Linezolid in the current standard of care for drug-resistant tuberculosis. However, a limitation of the work is the lack of control experiments with bedaquiline and pretomanid only, to further dissect the relevant contributions of linezolid and spectinamide in efficacy and adverse effects. Although the manuscript is well written overall, a re-formulation of some of the stated hypotheses and conclusions, as well as the addition of text to contextualize translatability, would improve its value.

Introduction

Tuberculosis (TB) remains one of the leading causes of death initiated by an infectious agent. In 2021, the World Health Organization reported 10.6 million new TB cases worldwide and among those, 450,000 cases were also diagnosed as multidrug resistant (MDR) or extensively drug resistant (XDR) TB (1 [↗](#)). Treatment of MDR- and XDR-TB patients is lengthy and is often poorly tolerated due to significant associated side effects (2 [↗](#)).

Recently, a 6-month novel treatment regimen of three-oral-drugs: bedaquiline (B), pretomanid (Pa) and linezolid (L) referred to as the BPaL regimen was approved. Preclinical studies demonstrated the better efficacy of BPaL regimen for drug sensitive TB compared to the standard TB chemotherapy (1 [↗](#), 3 [↗](#), 4 [↗](#)) and thereafter, the BPaL regimen was tested in the Nix-TB clinical trial conducted in South Africa (5 [↗](#)). This trial enrolled patients with XDR-TB and treatment-intolerant or non-responsive MDR-TB, including HIV positive patients with a CD4 count of 50 or higher. The results were remarkable as 95 out of 107 patients were cured though many patients had a high rate of treatment-associated adverse effects (AEs).

The long-term administration of linezolid (an oxazolidinone antibiotic) was likely the causative agent resulting in bone marrow myelosuppression (48%), peripheral neuropathy, optic neuritis (81%) and anemia (37%) in patients treated with the BPaL regimen (5 [↗](#)). Subsequently, the ZeNix trial adjusted the BPaL regimen to a linezolid dose of 600mg. This trial also had remarkable results; it cured 84-91% of patients (9 to 26 weeks of therapy, respectively) and resulted in fewer AEs than those observed in the Nix-TB trial (6 [↗](#)). Apart from AEs, several studies have also raised awareness of high doses of linezolid leading to development of linezolid-resistant *Mycobacterium tuberculosis* (Mtb) (7–9). At present, the TB-drug development field is working to modify the BPaL regimen to maintain or improve its efficacy while diminishing treatment-associated AEs (10 [↗](#)).

Spectinomycin, an aminocyclitol antibiotic, is a broad spectrum antibiotic used mainly for the treatment of *Neisseria gonorrhoeae* (11 [↗](#)). It inhibits bacterial protein synthesis (12 [↗](#)) and has a very high safety profile with no known ototoxicity and nephrotoxicity (13 [↗](#), 14 [↗](#)). The efficacy of spectinomycin against Mtb is very poor but its structural modification led to the development of a new series of semisynthetic analogs called spectinamides (12 [↗](#), 15 [↗](#), 16 [↗](#)). Spectinamides bind selectively with the bacterial 30S ribosomes and importantly, unlike linezolid, they do not bind to the mitochondrial 30S in mammalian cells. The latter represents a great advantage for reduced potential for side effects, such as ototoxicity and myeloid suppression that are commonly associated with other protein synthesis inhibitors such as amikacin and linezolid, respectively (17 [↗](#), 18 [↗](#)). Spectinamides' potent anti-tubercular activity is attributed to its ability to evade drug efflux by Rv1258c major facilitator superfamily transporter present on the surface of Mtb (19 [↗](#), 20 [↗](#)). Spectinamides have shown excellent activity against MDR- and XDR-Mtb strains (16 [↗](#), 21 [↗](#)). However, their poor oral availability have limited their usage to injectable forms.

One of the lead spectinamides, 1599, has demonstrated promising results *in vitro* and *in vivo* and was shown to lack cross-resistance with existing anti-TB drugs (16 [↗](#), 19 [↗](#)–23 [↗](#)). 1599, delivered subcutaneously, proved an effective partner agent when combined with rifampin and pyrazinamide and also with bedaquiline, pretomanid or moxifloxacin in TB mouse efficacy models of increasing complexity (21 [↗](#)).

One of the limitations of using 1599 as injectable is a potential risk for poor patient compliance (24 [↗](#)) and direct administration of aerosolized antibiotics to the lungs has been studied for decades as an alternative to systemic drug administration via injection. Aerosolized administration of 1599 has been tested in preclinical *in vivo* studies using liquid formulation of the drug. These studies have shown that inhaled 1599, used in monotherapy or in combination with pyrazinamide, is efficacious and well tolerated in murine TB efficacy models (25 [↗](#), 26 [↗](#)). A comparative study assessing biodistribution of the drug in relation to the administration route demonstrated that 1599 showed 48 times higher exposure in mouse lungs via inhalation compared to equivalent dosages administered by subcutaneous injection; the latter may explain the increased efficacy of this drug following intrapulmonary aerosol (22 [↗](#), 26 [↗](#)). Moreover, 1599 was shown to be amenable to dry powder formulation and delivery, suggesting a pathway to a more patient-friendly delivery system (27 [↗](#)). Therefore, in this study, we hypothesized that combining BPa with inhaled S will maintain equivalent efficacy to the BPaL regimen while avoiding the accompanying toxicities that occurred with long-term BPaL administration to human MDR/XDR-TB patients.

Results

Linezolid and spectinamide 1599 show similar efficacy in monotherapy

To compare the efficacy of L or S in monotherapy, Mtb-infected C3HeB/FeJ (n=7) and Balb/c (n=4-6) mice received L (administered 5/7 days per week orally at 100 mg/Kg) or S (administered 3/7 days per week on alternate days via intrapulmonary aerosol delivery at 100 mg/Kg and 50 mg/Kg, respectively) for 4 weeks. At the end of treatment, the animals were sacrificed, and the CFU in lungs and spleen was enumerated (**Figure 1A-D** [↗](#)). Treatment of C3HeB/FeJ mice with L ($7.30 \pm 0.45 \log_{10}$) or S ($6.90 \pm 0.48 \log_{10}$) for 4 weeks resulted in an average of 0.51 and 0.91 $\log_{10}$ CFU reduction in the lungs respectively, compared to untreated (UnRx) control ($7.81 \pm 0.36 \log_{10}$) (**Figure 1A** [↗](#)). This difference failed to achieve statistical significance owing to the larger standard deviation associated with heterogeneous advanced lung pathology observed in this model (26 [↗](#), 28 [↗](#), 29 [↗](#)). In contrast, L ($4.49 \pm 0.10 \log_{10}$) treated C3HeB/FeJ mice had significantly lower spleen bacterial burden compared to UnRx ($5.43 \pm 0.27 \log_{10}$), with no significant difference between the L and S ($4.94 \pm 0.11 \log_{10}$) treatment arm (**Figure 1C** [↗](#)). In Balb/c mice, L or S

treatment was found to promote significant reduction in lung bacterial burden (0.83 and $0.77 \log_{10}$ respectively) compared to UnRx or the vehicle only controls (**Figure 1B**). There was no significant difference in lung bacterial burden after L or S treatment in Balb/c mice and there was no change in spleen bacterial burden compared to UnRx control (**Figure 1D**).

Combination therapy with BPaL or BPaS have similar efficacy

We further tested and compared the efficacy of L or S when used in combination therapy with BPa. Mtb-infected C3HeB/FeJ mice were treated with either BPaL (B= 25mpk; Pa= 100 mg/Kg and L= 100 mg/Kg all administered 5/7 a week via oral gavage) or BPaS (BPa as in BPaL and S= 100 mg/Kg administered 3/7 a week on alternate days via intrapulmonary aerosol delivery) for 4 weeks. The comparative analysis from combined data of three independent studies is shown in **Figure 1E & G** (data from individual studies are shown in Figures S1-3). Two of three studies contained an extra group of BPa treated mice as a reference control. Compared to UnRx ($7.48 \pm 0.12 \log_{10}$) control, mice in the BPa ($5.49 \pm 0.42 \log_{10}$), BPaL ($4.76 \pm 0.38 \log_{10}$) and BPaS ($4.98 \pm 0.35 \log_{10}$) treatment groups had significantly reduced the lung bacterial burden by 1.99, 2.72, and 2.50 $\log_{10}$ respectively (**Figure 1E**). Although a higher CFU reduction was observed in the lungs of C3HeB/FeJ mice treated with BPaL or BPaS, these differences failed to achieve statistical significance compared to BPa backbone regimen. All three regimens proved highly effective at reducing spleen bacterial burden in C3HeB/FeJ mice, with most mice returning no CFU within the limit of detection (LOD) employed herein (**Figure 1G**).

The effect of BPaL and BPaS (S= 50 mg/Kg) combination therapy on the bacterial burden in lungs and spleen of Mtb-infected Balb/c mice was determined at the end of 4-weeks of treatment. **Figure 1F & H** show the combined lung and spleen CFU data from two independent studies (data for each study is shown in Figures S7-8). The combined result demonstrated that compared to UnRx ($5.24 \pm 0.17 \log_{10}$) control, mice in the BPaL ($1.29 \pm 0.13 \log_{10}$) and BPaS ($1.44 \pm 0.17 \log_{10}$) treatment groups returned significantly fewer CFU in the lungs, with most mice returning no CFU within the LOD employed herein (**Figure 1F**). As in C3HeB/FeJ TB model, no significant difference was observed in the lung CFU of Balb/c mice treated with either the BPaL or BPaS regimen. BPaL and BPaS therapy reduced Balb/c spleen bacterial burden to below the LOD of the assay with no CFU recovered for any treated mice (**Figure 1H**). In summary, these results support our hypothesis and demonstrate that both BPaL and BPaS multidrug regimens show equivalent bactericidal effects in C3HeB/FeJ and Balb/c chronic TB efficacy models.

Monitoring of adverse effects

Five approaches were employed to monitor treatment-associated AEs in mice in this study including 1) changes in the live body weight of mice; 2) lung histopathology and lesion scoring 3) evaluation of complete blood count (CBC), 4) clinical pathology to study myelosuppression in bone marrow and 5) changes in content of immune cells in the lungs, spleen, bone marrow and blood.

BPaL therapy decreases the live body weight of mice

No significant difference in the live body weight among the treatment groups in either C3HeB/FeJ or Balb/c mice was observed following 4-weeks of treatment with S or L alone (**Figure 2A & B**). On the other hand, C3HeB/FeJ or Balb/c mice treated with BPa, BPaL or BPaS showed marginal losses of live body weight, ranging from 2.37-5.13% and only mice receiving the BPaL regimen, when compared to the UnRx control, reached statistically significant losses in live body weight by the end of treatment (**Figure 2C & D**).

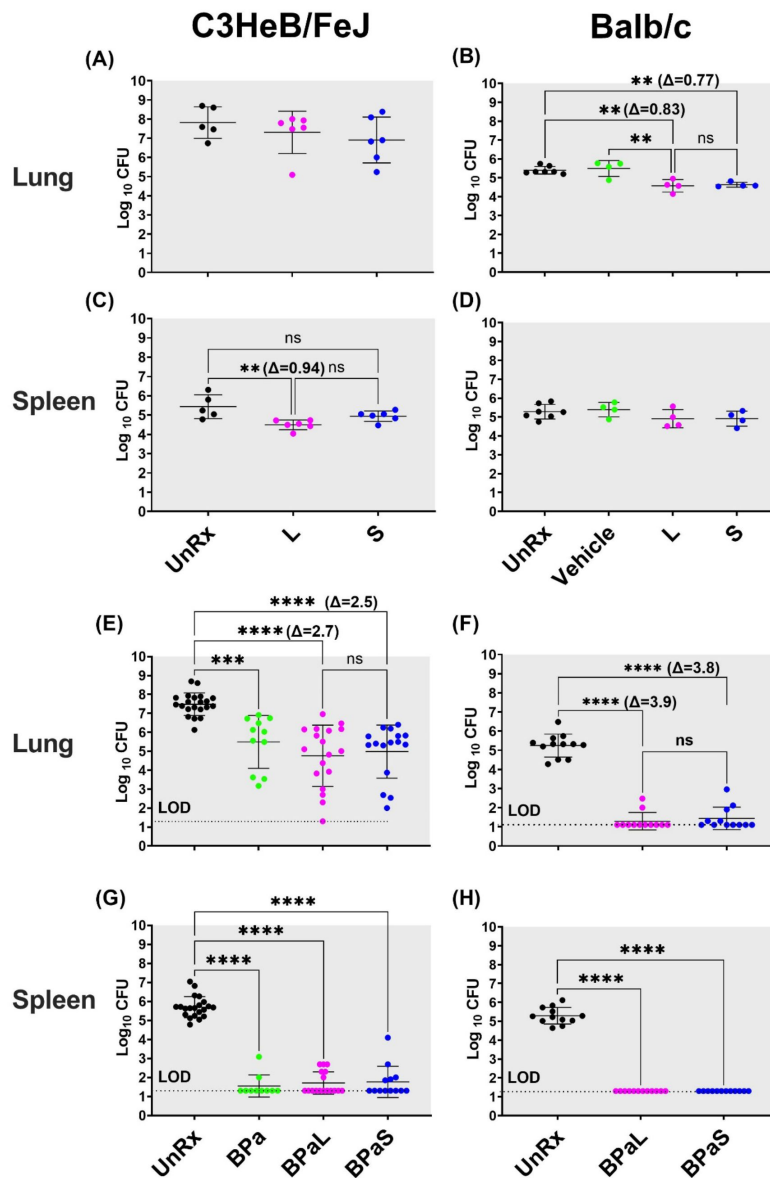


Figure 1.

Bactericidal effect of BPaL and BPaS in TB mouse models after 4-weeks of treatment

Balb/c and C3HeB/FeJ female mice were chronically infected with a low dose aerosol infection of Mtb Erdman strain to deliver ~75 and ~100 bacilli respectively. Post-infection, Balb/c and C3HeB/FeJ mice were rested for 4 and 8-9 weeks respectively until they were randomly assigned to the study groups. The mice were treated with monotherapy of linezolid or spectinamide 1599 or combination therapy of BPaL, BPa or BPaS for 4 weeks. Bedaquiline (B), pretomanid (Pa) and linezolid (L) were administered at 25, 100 and 100 mg/kg respectively by oral gavage for 5/7 a week while spectinamide 1599 at 50 and 100 mg/kg in Balb/c and C3HeB/FeJ TB models respectively for 3/7 a week on the alternate days via intrapulmonary aerosol delivery. On the third day of the last treatment, the mice were euthanized, and their lungs and spleen were collected. The organs were homogenized, serially diluted and plated on 7H11 agar with 4% charcoal (to avoid drug carry-over effect) to determine bacterial burden in the form of colony forming units (CFU) in each sample. CFU were enumerated after 4-6 weeks of incubation at 37 °C and expressed as log₁₀. C3HeB/FeJ and Balb/c TB models showing efficacy of monotherapy (A-D) and combination therapy (E-H). The C3HeB/FeJ graphs (E&G) represent the pooled data from three independent studies (n=3-8 each, Figures S1-3) and two of the three studies contained BPa as a reference control. The Balb/c graphs (F&H) represent the pooled data from two independent studies (n=5 each, Figures S7-8). Statistical significance was calculated using one-way ANOVA with Tukey's multiple comparison test, $p < 0.05$ was considered significant and ** = $p < 0.001$, *** = $p < 0.0001$, **** = $p < 0.0001$. UnRx = untreated, L = linezolid, S = spectinamide 1599, LOD: limit of detection.

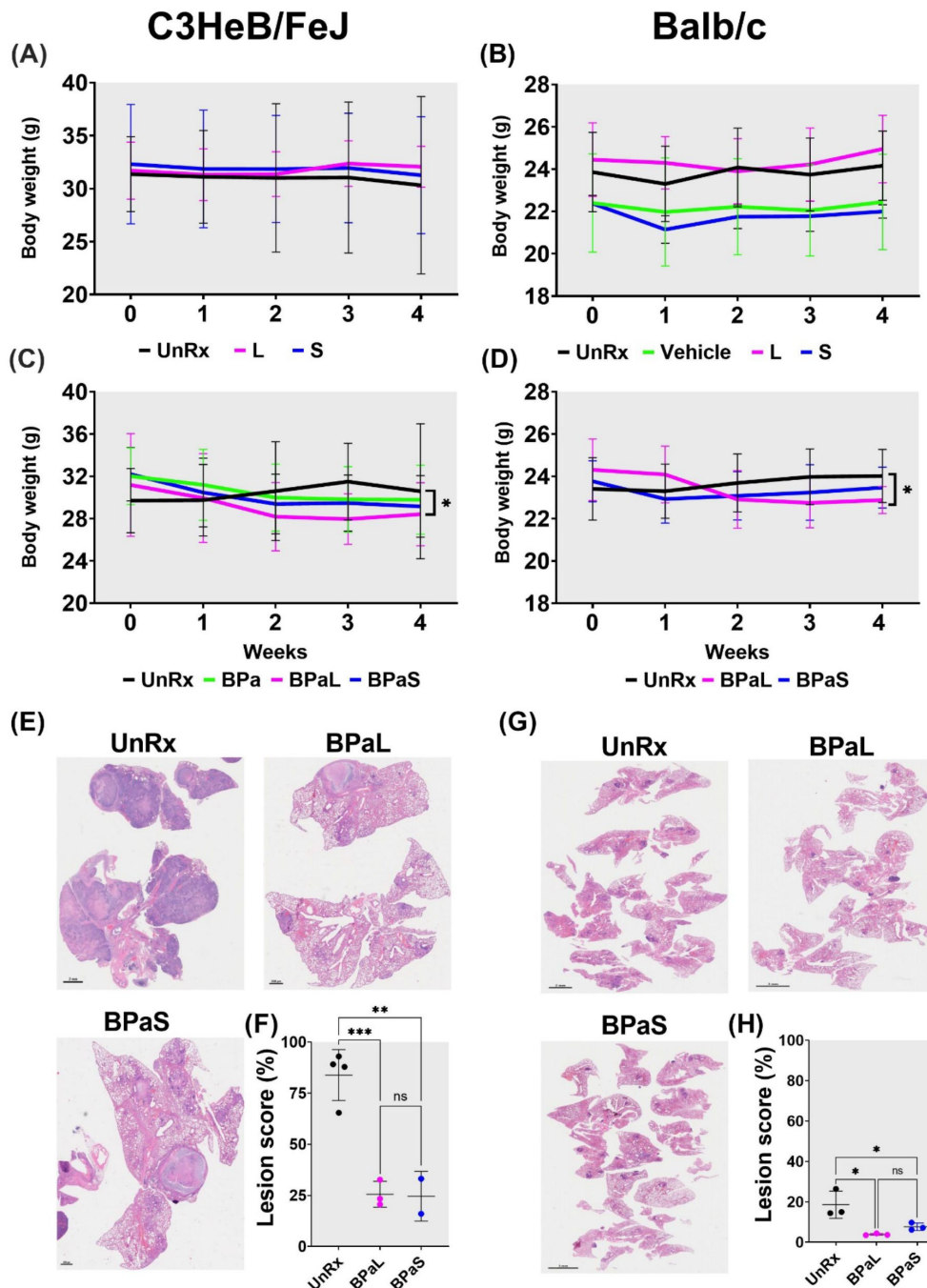


Figure 2.

Effect of therapy on the live body weight and lung histopathology of TB mouse models after 4-weeks of treatment

The live body weight of animals in [Figure 1](#) treated with monotherapy (A&B) or multidrug therapy (C&D) was recorded on a weekly basis. The average weight and SEM of each group is shown. At the end of therapy, the mice were euthanized, and their lungs were collected and processed for histopathology and lesion scoring (E-H). FFPE sections were cut at 5 μ m, stained with hematoxylin and eosin (H&E) and imaged at 40x (E&G). Lesion scores (F&H) were calculated as the proportion of infected area over the total lung area per animal. Statistical significance was calculated using mixed-effect model (body weight) and one-way ANOVA (lesion score) with Tukey's multiple comparison test, and $p < 0.05$ was considered significant and ** = $p < 0.001$, *** = $p < 0.0001$. UnRx = untreated, L = linezolid, S = spectinomamide 1599

BPaL and BPaS therapy result in significant lower lung lesion burden

The lesions in Mtb-infected UnRx C3HeB/FeJ mice (**Figure 2E**) showed a spectrum of diverse granuloma types ranging from aggregations of macrophages and lymphocytes to highly organized granulomas with collagen encapsulation and a region of central caseous necrosis that resembles to those found in some human patients (28,29). By comparison, lung lesions of Mtb-infected Balb/c mice (**Figure 2G**) consisted of granulomas with a very homogeneous structure (28) formed also by aggregations of macrophages and lymphocytes without a necrotic core. Mice treated for 4 weeks with the BPaL or BPaS regimen presented with significant reduction in the number and size of TB granulomas in both C3HeB/FeJ and Balb/c TB models compared to their respective UnRx control, with no significant difference in lesion burden score between the combination drug treatment groups (**Figure 2F & H**).

Association of L with altered blood profile and mild anemia in mice

Complete blood count (CBC) profile was performed on Mtb-infected C3HeB/FeJ mice at the end of 4-weeks of treatment to quantify treatment-associated hematological changes. The results obtained from mice treated with L or S alone are summarized in **Figure 3A**. Compared to UnRx control and S treated mice, L treatment significantly increased the red blood cell distribution width standard deviation (RDWs), while both L and S treatment were associated with a significant decrease in the mean corpuscular hemoglobin concentration (MCHC) compared to UnRx control (**Figure 3A**).

The effect of combination therapy with the BPaL or BPaS regimen on CBC profile was analyzed at 1-, 2- and 4-weeks of treatment (**Figure 3B**). None of the 20 parameters of CBC changed during the first 2 weeks of treatment. However, L-containing BPaL regimen was again associated with a significant increase in the RDWs, and lower hemoglobin (HGB) compared to UnRx control after 4 weeks. This effect was not observed in mice treated with BPa or BPaS (**Figure 3B**). However, as in monotherapy, there was a trend towards lower overall MCHC in mice treated with either BPaL or BPaS. The mean platelet volume (MPV) was marginally higher at 4 weeks in mice treated with BPa compared to BPaL or BPaS, albeit not significantly different from UnRx control (**Figure 3B**). Given that no difference in HGB or RDWs was observed between UnRx control and their comparator BPa and BPaS regimens, we concluded that the significant HGB decrease and RDWs increase (often observed during development of anemia) were associated with inclusion of L in BPaL regimen.

Spectinamide 1599 recovers the altered ratio of myeloid to erythroid cells in bone marrow

To further evaluate if L was associated with myelosuppressive effect, we performed hematopathology analysis on bone marrow from Mtb-infected C3HeB/FeJ and Balb/c mice at the end of treatment. For C3HeB/FeJ mice, the number of myeloid (M) and erythroid (E) precursor cells were calculated from H&E stained sections and their myeloid to erythroid ratio (M:E) was determined by counting 300 bone marrow cells in 5 different regions (**Figure 3C**). The BPa and BPaL treatment significantly decreased myeloid cells while increasing the proportion of erythroid cells (**Figure 3D**). Hence, the corresponding ratio in bone marrow of animals treated with BPaL or BPa was lower compared to UnRx control. Importantly, the BPaS treatment did not show any difference in the content of myeloid or erythroid cells when compared to UnRx control. In Mtb-infected Balb/c mice, the number of myeloid and other cell types were counted, however, no significant difference was found among the control and treatment groups (data not shown).

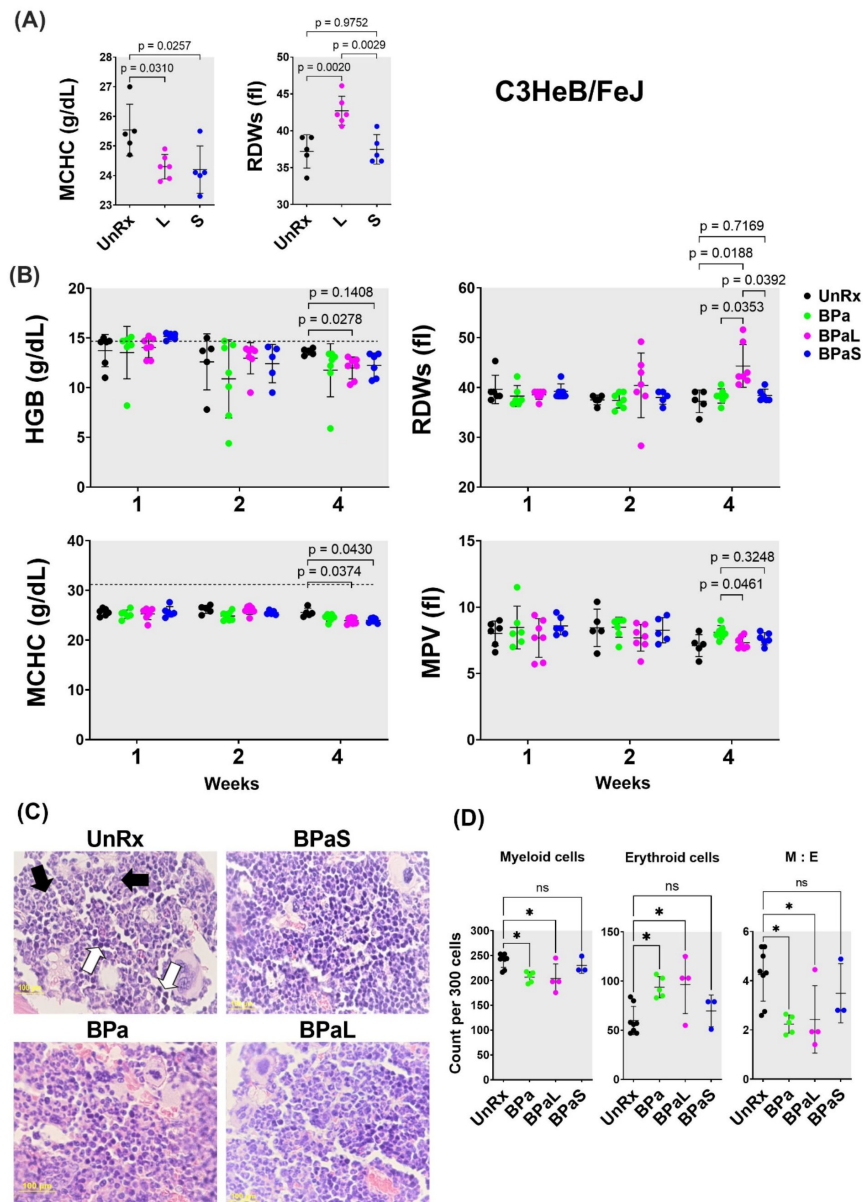


Figure 3.

Complete blood count profiling and bone marrow histopathology of Mtb infected C3HeB/FeJ TB model during 4-weeks of therapy

During therapy of mice in [Figure 1](#), the blood of live animals was collected at 1- and 2-weeks via the puncture of submandibular vein. At the end of 4-weeks therapy, the blood was collected by cardiac puncture. Blood was collected in K2-EDTA (purple top) tube and immediately analyzed for the complete blood count in VETSCAN® HM5 hematology analyzer (Zoetis). (A) monotherapy, (B) multidrug therapy. The MCHC (mean corpuscular hemoglobin concentration) and RDWs (red blood cell distribution width-standard deviation) along with the HGB (hemoglobin concentration) and MPV (mean platelet volume) are shown. A horizontal dotted line indicates the lower end of the reference interval for C3HeB/FeJ mice. The sternum, femur and tibiae bones from each mouse were collected, fixed in 4% PFA, and processed for histology (C&D). Sections were cut at 5 μ m, stained with hematoxylin and eosin (H&E) and imaged at 40x. (C) Representative photos of bone marrow sections showing myeloid (black arrows) and erythroid (white arrows) cells in bone marrow of untreated (UnRx) and treatment (BPa, BPaS and BPaS) groups. (D) The number of myeloid (M) and erythroid (E) among a total of 300 cells in 5 different regions were counted for each group and M:E was calculated. Statistical significance was calculated using one-way ANOVA with Tukey's test for multiple comparisons. p < 0.05 was considered significant and ** = p < 0.001, *** = P < 0.0001.

BPaL therapy increases proinflammatory cytokine response in bone marrow

A comparative analysis for concentration of cytokines and chemokines in the bone marrow, plasma, and lung samples from Mtb-infected C3HeB/FeJ mice treated with BPaL or BPaS regimen was also conducted. The bone marrow samples demonstrated a significant difference between the BPaL and BPaS groups, with appreciably higher level of pro-inflammatory cytokines and chemokines (IL-1 β , IL-12p70, IL-23, TNF α , GRO α (CXCL1), MP-2 α (CXCL2), IP-10 (CXCL10), MP-1 α (CCL3), RANTES (CCL5), MCP-3 (CCL7) and Eotaxin (CCL11)) in the BPaL-compared to BPaS-treated mice (**Figure 4B**). The plasma and lung samples, however, had similar cytokine and chemokine contents between the treatment groups except for MCP-3 (CCL7) in plasma which was significantly higher in BPaS compared to BPaL group (Figure S11). We also performed a correlation analysis of bone marrow cytokine and chemokine content with lung CFU obtained from treatment (BPaL or BPaS) and UnRx groups (**Figure 4C**). The analysis suggested that compared to UnRx control, there was a strong correlation between the pronounced reduction of lung bacterial burden and profound reduction in bone marrow cytokine and chemokine contents observed in mice from BPaL or BPaS groups. Similar correlations were found between lung CFU and content of cytokines and chemokines in lung (Figure S12) and plasma (Figure S13).

BPaL and BPaS therapies reduce inflammation-associated cells

We further assessed the environment of immune cells using flow cytometry in bone marrow, lungs, and blood from each group of Mtb-infected C3HeB/FeJ mice (**Figure 5**). The bone marrow (**Figure 5A**) results revealed that as compared to UnRx control, there was a significant decrease in the percentage of inflammatory myeloid phenotypes (CD45+CD3-CD11b+CD11c-Ly6C+CCR2+, CD45+CD3-CD11b+CD11c-Ly6C+CCR2+MHC-II+ and CD45+CD3-D14+CCR2+) in response to therapy with either BPa, BPaL, or BPaS. In contrast, neutrophils (CD45+CD3-CD11b+CD11c-Ly6C+Ly6G^{high}), precursor T cells (CD45+CD3+) and B cells (CD45+CD3-CD19+B220-) were significantly increased in either BPa, BPaL, or BPaS treatment groups compared to UnRx control in bone marrow. This reduced inflammatory response in treatment groups is also consistent in blood shown by significantly reduced inflammatory myeloid cells (CD45+CD3-CD11b+CD11c-Ly6C+CCR2+) (**Figure 5B**). Interestingly, the response to therapy in the lungs (**Figure 5C**) was manifested by a significant increase of CD3+CD4+ T helper cells and B-1 cells (CD3-CD19+) and a reverse trend for CD3+CD8+ and $\gamma\delta$ -T cells (CD3+CD8+ $\gamma\delta$ TCR+).

Furthermore, we also assessed changes in the distribution of immune cells in the lungs from Mtb-infected C3HeB/FeJ mice in response to therapy using multiplex fluorescent immunohistochemistry (mfiHC). A 7-color composite image for cell markers (B220, CD4, CD8, Foxp3, F4/80 and Ly6G) along with DAPI is shown in **Figure 6A**, while their single-color staining is shown in **Figure 6B**. **Figure 6A** shows a typical necrotic TB granuloma comprised of central necrosis, peripheral rim and lung parenchyma. The analysis of mfiHC images revealed that the BPaL and BPaS treatments significantly and dramatically lowered the number of neutrophils (count based on Ly6G+) compared to UnRx control, however, F4/80+ cells were observed significantly higher in BPaS compared to UnRx control (**Figure 6C**, S14). Interestingly, the spearman's correlation plot (**Figure 6D**) shows that a significant decrease in neutrophils was positively correlated with the corresponding increase in most of the other immune cells.

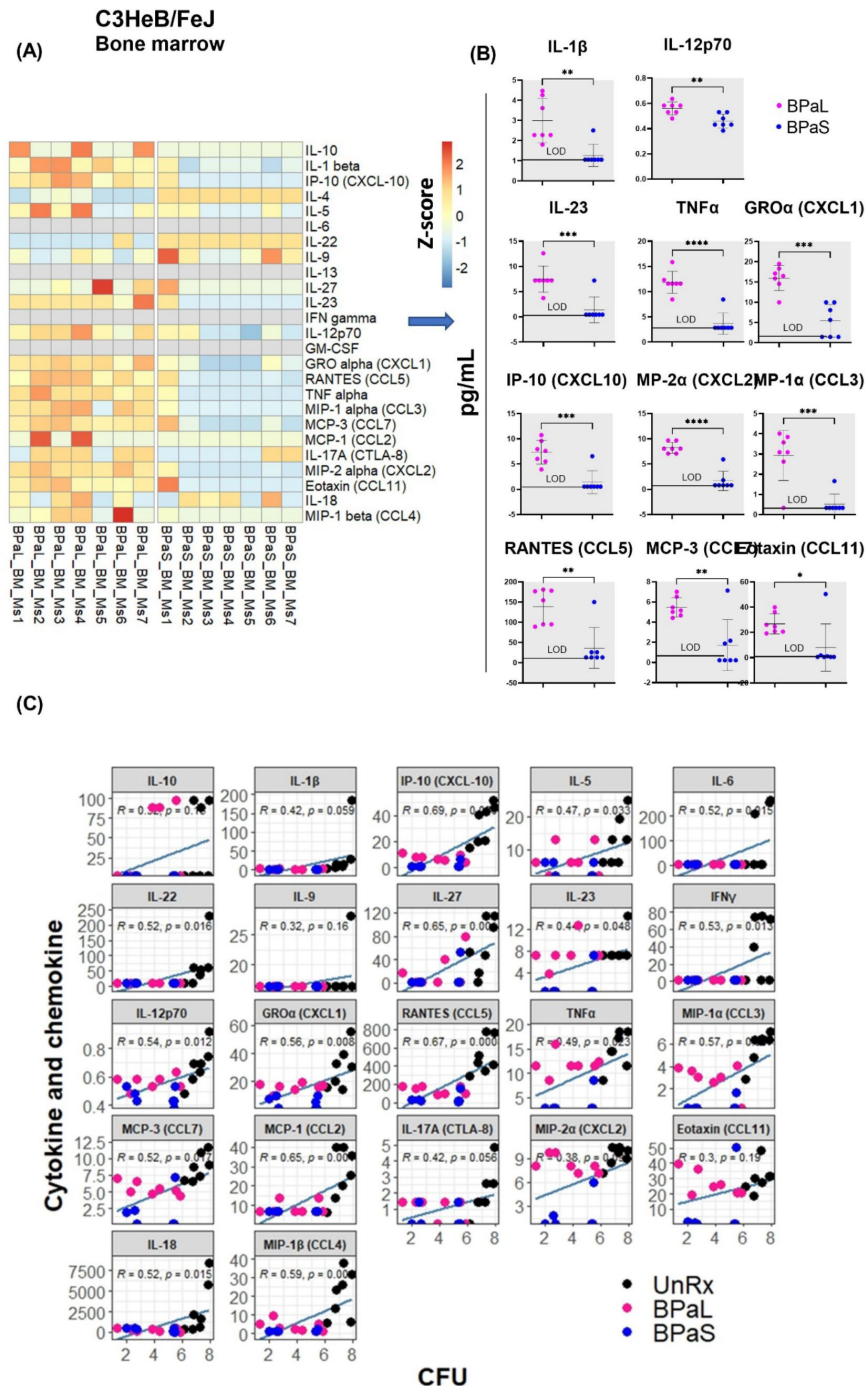


Figure 4.

Bone marrow cytokine and chemokine profiling in Mtb infected C3HeB/FeJ TB model after 4-weeks of treatment

Femur bones from selected studies in mice in [Figure 1](#) were collected to harvest the bone marrow. The bone marrow was resuspended in PBS, centrifuged and the supernatant was collected for the evaluation of cytokine's content. BPAL and BPAS therapy showed profile of 26 cytokines and chemokines in bone marrow and the data were converted to Z score and represented as a heatmap (A) and graphically (B). (C) Spearman's correlation analysis of bone marrow cytokines and chemokines with the lung bacterial burden (CFU).

Statistical significance was calculated using the t test. $p < 0.05$ was considered significant and ** = $p < 0.001$ *** = $p < 0.0001$, **** = $p < 0.00001$.

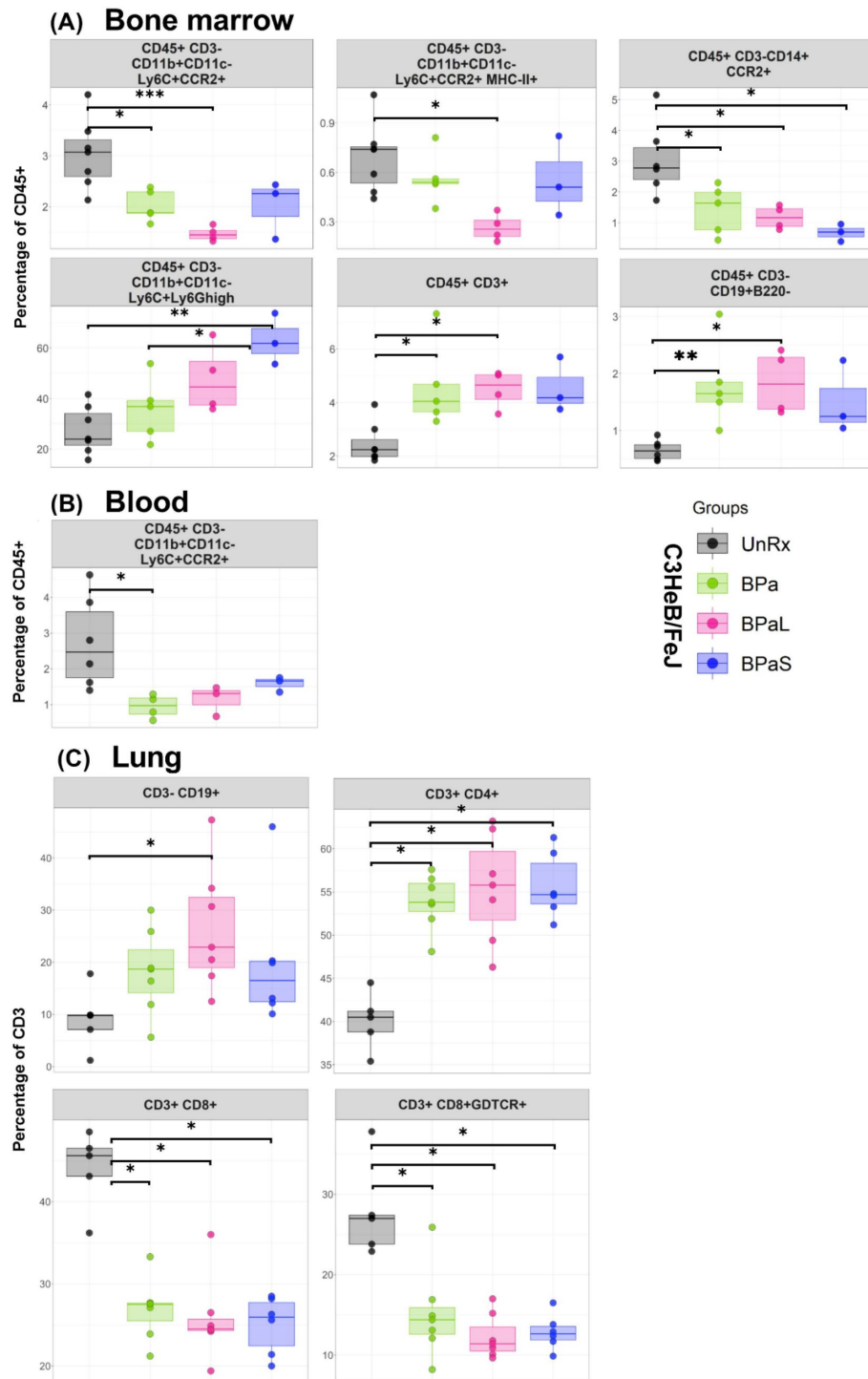


Figure 5.

Immune cell populations in the bone marrow, lung and blood of C3HeB/FeJ TB model after 4-weeks of treatment

The bone marrow, lung and blood from selected studies from [Figure 1](#) were evaluated by flow cytometry. The samples were processed for a panel of 17-color antibodies and the data were analyzed by FlowJo software using manual gating strategy. The myeloid and lymphoid phenotypes present in the untreated (UnRx) and treatment (BPa, BPaL or BPaS) groups are shown. Statistical significance was calculated using one-way ANOVA with Tukey's test for multiple comparisons. $p < 0.05$ was considered significant and ** = $p < 0.001$ *** = $p < 0.0001$, **** = $p < 0.00001$.

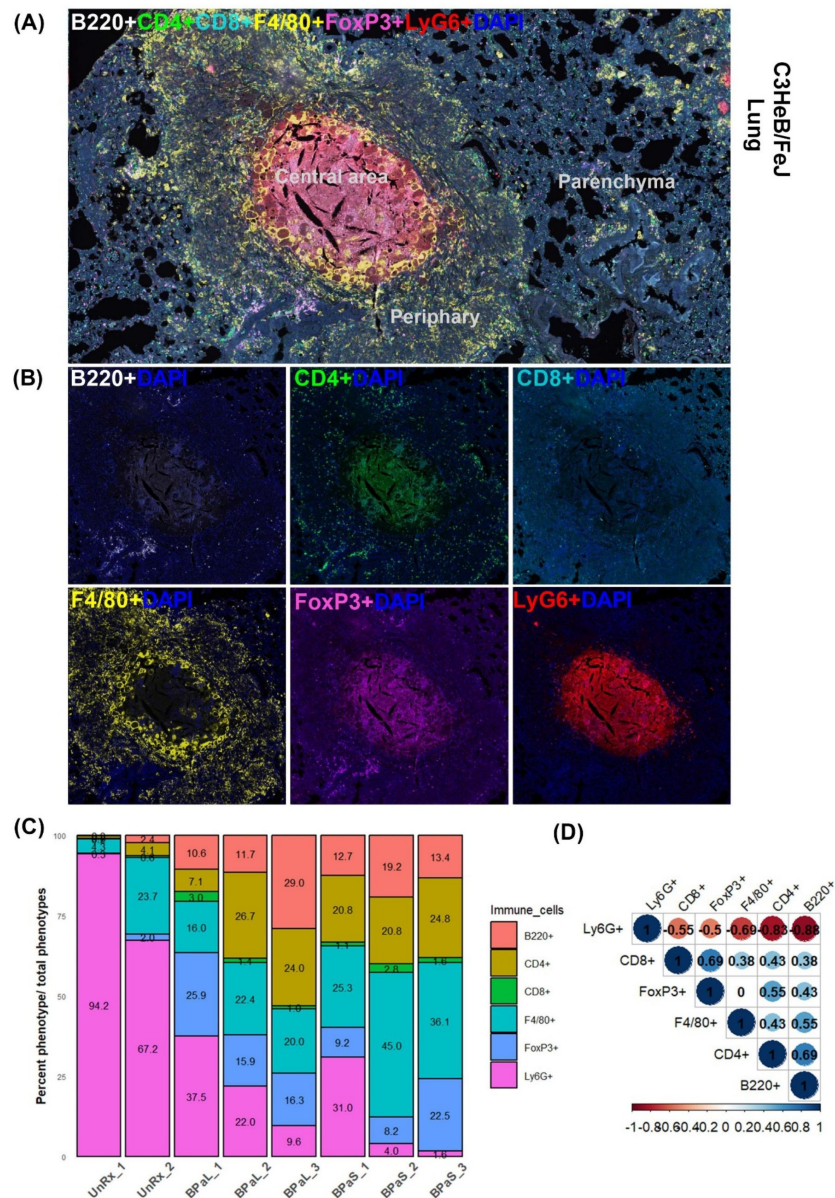


Figure 6.

Immune cell populations in the lungs of Mtb infected C3HeB/Fej TB model after 4-weeks of treatment

Selected mice from those shown in [Figure 1](#) were processed for multiplex fluorescent immunohistochemistry (mfiHC). The mfiHC was performed for a panel of 6-color antibodies + DAPI using Opal-plex Tyramide Signal Amplification (TSA). Slides were scanned using multispectral automated PhenoImager™ (Akoya Biosciences) and analyzed for different immune cell populations using the inForm® tissue Finder and Phenochart software (Akoya Biosciences). (A) The lung mfiHC full composite image displays B220, CD4, CD8, F4/80, FoxP3 and LyG6 markers along with DAPI staining for nuclei in the TB granuloma. The central and peripheral regions of a TB granuloma and the parenchyma of lung are also shown. (B) Single color composite image of individual markers with DAPI showing distribution of each immune cell population in the TB granuloma. (C) Cell populations (%) of several immune cells per total number of phenotypes calculated in untreated (UnRx) and treatment (BP aL and BP aS) groups based on a panel of 6-color antibodies + DAPI. (D) Spearman's correlation matrix for several immune cell populations (B220, CD4, CD8, F4/80, FoxP3, LyG6) showing all relationships. A coefficient with a value of either +1 (blue), 0 (white), or -1 (red) indicates a perfect association, no association, and a perfect negative association of ranks, respectively. Numbers indicate the correlation coefficient.

Discussion

We used two preclinical chronic TB murine efficacy models (30–32) to investigate BP_a, BP_aL and BP_aS, for efficacy and any associated AEs during the course of 4-weeks of treatment. Based on the diversity of outcomes observed during human TB disease (22) and as no single animal model recapitulates the wide spectrum of human TB pathology (33), we chose the Balb/c and C3HeB/FeJ murine TB models. The Balb/c chronic TB model is representative of a long-term Mtb chronic infection that develops homogenous lung granulomatous lesions restraining the bacilli within intracellular compartments (34) of macrophages and foamy macrophages. In contrast, low-dose aerosol Mtb infection of C3HeB/FeJ mice also results in a chronic infection, but their lungs exhibit a heterogenous spectrum of lesions including granulomas similar to those seen in Balb/c chronic TB model in addition to caseous necrotic lesions surrounded by a fibrotic rim (28,29,35). The caseum of these necrotic granulomas creates a hypoxic environment and contains abundant extracellular bacilli (28,29,36) in a similar fashion to necrotic granulomas found in some human TB patients. It is believed that the fibrotic, necrotic and hypoxic environment of these granulomas creates barriers to drug penetration, alters bacterial phenotype, and all together challenge therapeutic outcomes (30–32,37). Therefore, to understand the implications of drug efficacy and drug associated AEs in scenario without (Balb/c) and with (C3HeB/FeJ) necrotic granulomas, both murine TB efficacy models were employed.

Overall, our antimicrobial data are in accordance with the granuloma spectrum of both mouse models discussed above where a more robust reduction in lung bacterial burden was observed in the absence (Balb/c) versus the presence (C3HeB/FeJ) of necrotic lesions. Both multidrug regimens as BP_aL or BP_aS significantly reduced lung bacterial burden by 3.8–4.0 log₁₀ and 2.5–2.7 log₁₀ in Balb/c and C3HeB/FeJ TB models, respectively (Figures 1E–H). We therefore conclude that both regimens promote similar bactericidal effects in murine models lacking, or featuring, advanced pulmonary pathology. Furthermore, the potent antimicrobial effect of the BP_aL and BP_aS regimens also resulted in improvement of the pathological outcome during chronic infection with Mtb but only the L-containing BP_aL regimen, not BP_aS, was associated with a significant decrease in the live body weight at week 4 in Mtb-infected Balb/c and C3HeB/FeJ mice (Figure 2C & D). The extent of weight loss is an important preclinical and clinical parameter in TB patients because it determines the severity of disease progression (38), and it is also an indicator of *in vivo* drug efficacy (39). The bactericidal effects of S (as monotherapy) (40) and BP_aL (41) in mice observed in these studies is in line with previous reports and similar inverse relationship between body weight and L exposure was recently reported in human patients (42).

Among L-associated hematological side effects, the incidence of anemia is reported up to 62.5% in MDR and XDR-TB patients (43–45) and the onset of this effect can occur 2 weeks to 2 months post L administration (46). In human TB patients treated with anti-TB drugs (47–49), increased RDWs and lower HGB are associated with anemia and these parameters serve as markers of disease prognosis. Using a CBC profile of 20 peripheral blood parameters, our preclinical study failed to detect any difference between L or S 4-weeks monotherapy in terms of total red blood cells, white blood cells, platelets, or hemoglobin (HGB) concentration compared to UnRx control (data not shown). However, L treatment alone increased the RDWs and both L and S decreased the MCHC (Figure 3A). Likewise, none of the 20-blood parameters evaluated showed any change after the first 2 weeks of therapy with BP_a, BP_aL, or BP_aS (Figure 3B). However, by 4 weeks, a significant drop in HGB and an increase in the RDWs was apparent for the BP_aL group (Figure 3B). This is interpreted to mean that mild hematological effects observed in mice treated for 4 weeks with L or BP_aL are dependent on the number of L-doses administered and are thus, time-dependent. Overall, the onset of these hematological effects when testing an L-containing regimen (BP_aL) is in agreement with previous studies (46,50) in human patients.

The mechanism of L-induced toxicity is attributed to its binding with the host mitochondrial ribosomes leading to mitochondrial toxicities (17). The latter results in activation of Nlrp3 inflammasome (51) and subsequently results in L-mediated bone marrow myelosuppression (52); a phenomenon consistent with the hematologic anomalies seen in patients treated with L for extended time periods. Because spectinamides do not bind to mitochondrial ribosomes there is reduced potential for similar side effects (16,18). To conclusively test this hypothesis, we performed bone marrow histopathology to quantify the myeloid to erythroid ratio (M:E), a parameter that provides information about the relative proportions of myeloid lineage (granulocytes, monocytes, and their precursors) to erythroid lineage (53). The BPa and BPaL regimens altered M:E in the C3HeB/FeJ TB model by suppressing myeloid and inducing erythroid lineages (Figure 3C&D) whereas no such difference was observed in mice treated with BPaS compared to untreated control. L was previously shown to impact M:E ratio, although an opposite trend was observed in those studies, which employed 12-days of L administration and a different strain of otherwise healthy mice (51). Time course studies using a single consistent assay method are needed to resolve this discrepancy.

Elevation of interleukin 1 β (IL-1 β) levels and activation of Nlrp3 inflammasome have been previously linked to myelosuppression (52). A 26-plexed immunoassay on bone marrow samples from Mtb-infected C3HeB/FeJ mice revealed that most of the proinflammatory cytokines and chemokines including IL-1 β , IL-12p70 and TNF- α were present at significantly higher concentrations in animals treated with BPaL compared to BPaS (Figure 4A&B). The presence of elevated IL-1 β was previously reported during monotherapy with L (51,54). Studies from UnRx Mtb-infected C3HeB/FeJ mice also revealed highly elevated levels of cytokines and chemokines (Figure 4C). As expected, the positive therapeutic effect of BPaL and BPaS (as seen by reduction in bacterial and lesion burden of lungs) also correlated with decreased level of proinflammatory cytokines in bone marrow, lung, and plasma (Figures 4C, S12 and S13 respectively).

Similar to changes observed for cytokine profile, the flow cytometry data of bone marrow, blood, and lungs along with quantification of mFIHC lung image analysis revealed a significant change in the percentage of myeloid and lymphoid phenotypes during treatment compared to UnRx control (Figures 5 and 6). Most notably, the therapeutic effect of the BPa, BPaL and BPaS treatments reduced inflammatory myeloid cells expressing CCR2 in blood and bone marrow and reduction in cytotoxic T cells (CD3+CD8+) and $\gamma\delta$ T cells (CD3+CD8+ $\gamma\delta$ TCR+) in lungs. In addition, BPa, BPaL and BPaS therapy significantly increased the influx of helper T cells (CD3+CD4+), regulatory T cells (Foxp3+) and B cells (CD3-CD19+) in lungs. These results suggest that the combination therapy promotes the immune system's equilibrium by reducing inflammation and enhancing adaptive immune responses.

An additional striking finding from this study was a strong concordance between the decrease in lung and spleen bacterial burden and a corresponding decrease in the number of cells expressing the neutrophil associated marker (Ly6G) (Figure 6C&D). An implication of this finding is that the favorable treatment outcomes may promote a corresponding decline in Ly6G neutrophil population as suggested before (55). Among all immune cell phenotypes studied, no differences were found between BPaL and BPaS treated animals and only the F4/80 +cells in the BPaS (but not in the BPaL) treated animals showed significant increase when compared to the UnRx control.

The TB drug development field is working towards developing shorter and safer therapies with a common goal of developing new multidrug regimens of low pill burden that are accessible to patients, of short duration (ideally 2-3 months) and consist of 3-4 drugs of novel mode-of-action with proven efficacy, safety, and limited toxicity. Here we present initial results that are in line with these goals. Further studies with inhaled spectinamide 1599 and BPa are therefore warranted to determine if 2-3 months of treatment with BPaS, further reduces the bacterial burden and leads to sterilizing activity without associated AEs.

Materials and Methods

Female C3HeB/FeJ and Balb/c mice at 6-8 weeks of age were purchased from the Jackson Laboratories. All protocols and use of these animals were approved by the Institutional Animal Care and Use Committee (IACUC) at CSU. Animals were infected with a low dose aerosol infection of Mtb (Erdman strain; ATCC 35801) using an inhalation exposure system (Glas-Col, Terre Haute, IN) calibrated to deliver ~50-100 colony forming units (CFU) to the lungs (22 [↗](#)). Clinical observations (e.g., inactivity, rough fur, hunched posture, increased respiratory rate or effort) were monitored daily and their live body weights were taken weekly.

Drug preparation and treatment

Bedaquiline fumarate (B) and linezolid (L) were obtained from LKT laboratories, pretomanid (Pa) from ChemShuttle and 1599 (S: dihydrochloride) was provided by Dr. Lee at St. Jude Children's Research Hospital. B was administered at 25 mg/kg. Pa and L were administered at 100 mg/kg each and S was delivered by inhalation in liquid form at 50 and 100 mg/kg to Balb/c and C3HeB/FeJ mice, respectively. The drugs were formulated in weekly batches according to the live body weight of the animals, aliquoted for single daily dosing, and stored at 4 °C in the dark.

The drugs were prepared as reported previously (21 [↗](#)). Drug treatment was started 4 weeks post-aerosol infection for Balb/c and at 8-9 weeks post-aerosol infection for C3HeB/FeJ mice to allow time for lung pathology to fully develop. All drugs were administered once daily for 5 days/week for 4 weeks by oral (gavage) administration except S which was administered 3 days/week by intrapulmonary aerosol delivery using the Penn Century microsyringe as reported previously (22 [↗](#)). B was administered in the morning, Pa one hour after B and, L and S at least 4 hours after Pa.

Necropsy

After 4-weeks of treatment, C3HeB/FeJ and Balb/c mice were humanely euthanized by CO₂ narcosis. Blood, lungs, spleen, femur and tibia bones were collected from each mouse for further processing and analysis.

Assessment of Efficacy

The efficacy of the treatment was assessed by determining changes in bacterial burden [measured as colony forming units (CFU)] in the lungs and spleen of animals at necropsy. The lungs and spleen were homogenized and prepared as reported previously (22 [↗](#)). The lung homogenates were plated onto 7H11 agar plates supplemented with 0.4% activated charcoal to reduce the carryover effect of drugs and incubated at 37 °C for 6-8 weeks before the final CFU count. The remaining lung homogenate was centrifuged, and the supernatant was collected and stored at -80 °C for evaluation of cytokines and chemokines.

Histopathology and lesion scoring

The lungs were fixed in 4% paraformaldehyde (PFA) for 48 hours then embedded in paraffin for histopathology. Sections from formalin fixed and paraffin embedded (FFPE) tissues were cut at 5 µm, stained with hematoxylin and eosin (H&E) and scanned at 40X magnification using multispectral automated PhenoImager (Akoya Biosciences) for histopathological evaluation. The extent of lung lesion burden was quantified in blinded digital images using an open-source QuPath software for image analysis as described previously (56 [↗](#)).

The sternum and one femur were fixed in 4% PFA and processed for histology. To evaluate the myelosuppressive effect of the drugs, bone sections were cut at 5 μm and stained with H&E. The number of myeloid and erythroid cells from 5 different regions of the bone were blinded and then counted by a veterinary clinical pathologist.

Bone marrow collection

Briefly, a 0.6 mL sterile Eppendorf tube punctured at the bottom with the help of a 26-gauge needle was inserted into a 1.5 mL sterile Eppendorf tube. One end of the epiphysis of the long bones was cut open to expose the bone marrow and placed down into the small Eppendorf tube-system. The tubes were centrifuged at 10,000 $\times g$ for 15 seconds and the marrow was collected from the base of the large Eppendorf tube. The bone marrow was resuspended in PBS and centrifuged again. Thereafter, the supernatant was collected and stored at -80°C for evaluation of cytokines and chemokines while the bone marrow cells were saved in 4% PFA and freezing media for further use in clinical pathology analysis and flow cytometry, respectively.

Processing of blood

For complete blood count (CBC) analysis of C3HeB/FeJ animals during the treatment, blood was collected in K2-EDTA tubes via submandibular vein puncture as described previously (57). The blood was immediately analyzed in VETSCAN[®] HM5 hematology analyzer (Zoetis).

At the time of necropsy, whole blood was collected via cardiac puncture in K2-EDTA containing tubes. After adding an equal volume of PBS, the samples were centrifuged at 800 $\times g$ for 10 min at 25°C with the brake off (deceleration = 0). The top plasma layer was collected and stored at -80°C for evaluation of their cytokine's content. The buffy coat was collected, washed and the erythrocytes were lysed using Miltenyi RBC lysis buffer (Miltenyi, CA). The cells were washed and resuspended in 500 μL of complete DMEM media and prepared for flow cytometry analysis.

Cytokine quantification

Multiplex immunoassay was performed using a Luminex bead-based multiplex ELISA kit (ProcartaPlex Mouse Cytokine & Chemokine Panel 1 26plex, reference # EPXR260-26088-901, Invitrogen). Each sample was normalized to the total protein concentration determined by Bicinchoninic acid (BCA) assay (Thermo Fisher). The BCA and Luminex assay were performed according to the manufacturer's instructions and the final stained samples were fixed with 4% PFA prior to acquisition. Sample data were acquired on a MAGPIX instrument running xPONENT 4.3 software (Luminex Corp.). Heatmaps were generated using R pheatmap package. Correlation analysis of cytokine contents in bone marrow, plasma and lungs with the lung bacterial burden was performed using the corrplot package in R.

Immune cell populations analysis

Single cell suspension of bone marrow, blood, and lungs from C3HeB/FeJ mice was prepared as described previously (56). Cells counting, viability staining and cell staining (Table S1) was performed accordingly (56). Samples were acquired using Cytex Aurora[™] 4-Laser spectral flow cytometer where 100,000 events were recorded. Data were analyzed in FlowJo software (BD Biosciences) using manual gating (58).

Multiplex fluorescent immunohistochemistry

Five μm sections of FFPE lung tissues were stained for multiplex fluorescent immunohistochemistry (mfiHC) by the Imaging Core at the University of Colorado, Anschutz Medical Campus, Denver. The mfiHC was performed for a panel of 6-color antibodies + DAPI using Opal-plex Tyramide Signal Amplification (TSA) technique using a Leica Bond III autostainer. The details of antibodies and Opal fluorophores used are given in Table S2. Each antibody was

optimized using Opal 3-Plex Anti-Rb Detection Kit (Akoya Biosciences Inc. cat# NEL830001KT) and stained with automated LabSatTM Research (Lunaphore Technologies SA, Eprexia). Slides were scanned using multispectral automated PhenoImagerTM (Akoya Biosciences) and analyzed for several immune cell populations using the inForm[®] tissue Finder (Version 2.4.8) and PhenochartTM (Version 1.0.12) software (Akoya Biosciences).

Statistical analysis

Bacterial burden data were expressed as CFU which were Log₁₀-transformed and analyzed using GraphPad Prism version 9.5.1 (GraphPad software, La Jolla, CA). The statistical analysis was performed using a Tukey's multiple comparison test as part of either one-way or two-way ANOVA and mixed-model effect where necessary. The correlation analysis was performed using the spearman's correlation test. Flow cytometry and mFIHC data were graphed in R studio and statistical evaluation was performed using stats package in R.

Acknowledgements

This research was supported by the National Institute of Allergy and Infectious Diseases and the Office of the Director of the National Institutes of Health (grants R01AI120670, R01AI090810 and 1S10OD030263 respectively). Dr. Malik Zohaib Ali was a Fulbright Foreign Student Program grantee from Pakistan. Dr. Charles Daley from the Division of Mycobacterial and Respiratory Infections, National Jewish Health, Denver, Colorado 80206, USA advised on current TB drug treatments. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health. We acknowledge the staff of the Laboratory Animal Resources at Colorado State University for providing animal care.

Supporting information

Figure S1 (C3HeB/FeJ Study 1). Bacterial burden (CFU) in Mtb infected C3HeB/FeJ mice treated with BPAL and BPAS regimen for 4 weeks

Figure S2 (C3HeB/FeJ Study 2). Bacterial burden (CFU) in Mtb infected C3HeB/FeJ mice treated with BPA, BPAL and BPAS regimen for 4 weeks

Figure S3 (C3HeB/FeJ Study 3). Bacterial burden (CFU) in Mtb infected C3HeB/FeJ mice treated with BPA, BPAL and BPAS regimen for 4 weeks

Figure S4 (C3HeB/FeJ Study 1). Change in the average live body weight of Mtb infected C3HeB/FeJ mice during drug treatment

Figure S5 (C3HeB/FeJ Study 2). Change in the average live body weight of Mtb infected C3HeB/FeJ mice during drug treatment

Figure S6 (C3HeB/FeJ Study 3). Change in the average live body weight of Mtb infected C3HeB/FeJ mice during drug treatment

Figure S7 (Balb/c Study 1). Bacterial burden (CFU) in Mtb infected Balb/c mice treated with BPAL and BPAS regimen for 4 weeks

Figure S8 (Balb/c Study 2). Bacterial burden (CFU) in Mtb infected Balb/c mice treated with BPAL and BPAS regimen for 4 weeks

Figure S9 (Balb/c Study 1). Change in the average live body weight of Mtb infected Balb/c mice during drug treatment

Figure S10 (Balb/c Study 2). Change in the average live body weight of Mtb infected Balb/c mice during drug treatment

Figure S11. Change in the cytokines and chemokines profile in Mtb infected C3HeB/FeJ mice during drug treatment

Figure S12. Spearman's correlation analysis between bacterial burden (CFU) and cytokine and chemokine profile in the lungs of Mtb infected C3HeB/FeJ mice treated with BPaL and BPaS regimen

Figure S13. Spearman's correlation analysis between lung bacterial burden (CFU) and plasma cytokine and chemokine profile in Mtb infected C3HeB/FeJ mice treated with BPaL and BPaS regimen

Figure S14. Immune cell populations in the lungs of Mtb infected C3HeB/FeJ TB model after 4 weeks therapy

Table S1. Reagents

Table S2. Antibodies

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Reviewer #1 (Public Review):

Summary:

This manuscript is an extension of previous studies by this group looking at the new drug spectinamide 1599. The authors directly compare therapy with BPaL (bedaquiline, pretomanid, linezolid) to a therapy that substitutes spectinamide for linezolid (BPaS). The Spectinamide is given by aerosol exposure and the BPaS therapy is shown to be as effective as BPaL without adverse effects. The work is rigorously performed and analyses of the immune responses are consistent with curative therapy.

Strengths:

- (1) This group uses 2 different mouse models to show the effectiveness of the BPaS treatment.
- (2) Impressively the group demonstrates immunological correlates associated with Mtb cure with the BPaS therapy.
- (3) Linezolid is known to inhibit ribosomes and mitochondria whereas spectinamide does not. The authors clearly demonstrate the lack of adverse effects of BPaS compared to BPaL.

Weaknesses:

- (1) Although this is not a weakness of this paper, a sentence describing how the spectinamide would be administered by aerosolization in humans would be welcomed.

<https://doi.org/10.7554/eLife.96190.1.sa2>

Reviewer #2 (Public Review):

Summary:

Replacing linezolid (L) with the preclinical development candidate spectinamide 1599, administered by inhalation, in the BPaL standard of care regimen achieves similar efficacy, and reduces hematological changes and pro-inflammatory responses.

Strengths:

The authors not only measure efficacy but also quantify histological changes, hematological responses, and immune responses, to provide a comprehensive picture of treatment response and the benefits of the L to S substitution.

The authors generate all data in two mouse models of TB infection, each reproducing different aspects of human histopathology.

Extensive supplementary figures ensure transparency.

Weaknesses:

The articulation of objectives and hypotheses could be improved.

<https://doi.org/10.7554/eLife.96190.1.sa1>

Reviewer #3 (Public Review):

Summary:

In this paper, the authors sought to evaluate whether the novel TB drug candidate, spectinamide 1599 (S), given via inhalation to mouse TB models, and combined with the drugs B (bedaquiline) and Pa (pretomanid), would demonstrate similar efficacy to that of BPaL regimen (where L is linezolid). Because L is associated with adverse events when given

to patients long-term, and one of those is associated with myelosuppression (bone marrow toxicity) the authors also sought to assess blood parameters, effects on bone marrow, immune parameters/cell effects following treatment of mice with BPaS and BPaL. They conclude that BPaL and BPaS have equivalent efficacy in both TB models used and that BPaL resulted in weight loss and anemia (whereas BPaL did not) under the conditions tested, as well as effects on bone marrow.

Strengths:

The authors used two mouse models of TB that are representative of different aspects of TB in patients (which they describe well), intending to present a fuller picture of the activity of the tested drug combinations. They conducted a large body of work in these infected mice to evaluate efficacy and also to survey a wide range of parameters that could inform the effect of the treatments on bone marrow and on the immune system. The inclusion of BPa controls (in most studies) and also untreated groups led to a large amount of useful data that has been collected for the mouse models per se (untreated) as well as for BPa - in addition to the BPaS and BPaL combinations which are of particular interest to the authors. Many of these findings related to BPa, BPaL, untreated groups, etc corroborate earlier findings and the authors point this out effectively and clearly in their manuscript. To go further, in general, it is a well-written and cited article with an informative introduction.

Weaknesses:

The authors performed a large amount of work with the drugs given at the doses and dosing intervals started, but at present, there is no exposure data available in the paper. It would be of great value to understand the exposures achieved in plasma at least (and in the lung if more relevant for S) in order to better understand how these relate to clinical exposures that are observed at marketed doses for B, Pa, and L as well as to understand the exposure achieved at the doses being evaluated for S. If available as historical data this could be included/cited. Considering the great attempts made to evaluate parameters that are relevant to clinical adverse events, it would add value to understand what exposures of drug effects such as anemia, weight loss, and bone marrow effects, are being observed.

It would also be of value to add an assessment of whether the weight loss, anemia, or bone marrow effects observed for BPaL are considered adverse, and the extent to which we can translate these effects from mouse to patient (i.e. what are the limitations of these assessments made in a mouse study?). For example, is the small weight loss seen as significant, or is it reversible? Is the magnitude of the changes in blood parameters similar to the parameters seen in patients given L?

In addition, it is always challenging to interpret findings for combinations of drugs, so the addition of language to explain this would add value: for example, how confident can we be that the weight loss seen for only the BPaL group is due to L as opposed to a PK interaction leading to an elevated exposure and weight loss due to B or Pa?

Turning to the evaluations of activity in mouse TB models, unfortunately, the evaluations of activity in the BALB/c mouse model as well as the spleens of the Kramnik model resulted in CFU below/at the limit of detection and so, to this reviewer's understanding of the data, comparisons between BPaL and BPaS cannot be made and so the conclusion of equivalent efficacy in BALB/c is not supported with the data shown. There is no BPa control in the BALB/c study, therefore it is not possible to discern whether L or S contributed to the activity of BPaL or BPaS; it is possible that BPa would have shown the same efficacy as the 3 drug combinations. It would be valuable to conduct a study including a BPa control and with a shorter treatment time to allow comparison of BPa, BPaS, and BPaL. In the Kramnik lungs, as the authors rightly note, the studies do not support any contribution of S or L to BPa - i.e. the activity observed for BPa, BPaL, and BPaS did not significantly differ. Although the

conclusions note equivalency of BPaL and BPaS, which is correct, it would be helpful to also include BPa in this statement; it would be useful to conduct a study dosing for a longer period of time or assessing a relapse endpoint, where it is possible that a contribution of L and/or S may be seen - thus making a stronger argument for S contributing an equivalent efficacy to L. The same is true for the assessment of lesions - unfortunately, there was no BPa control meaning that even where equivalency is seen for BPaL and BPaS, the reader is unable to deduce whether L or S made a contribution to this activity.

<https://doi.org/10.7554/eLife.96190.1.sa0>